A Reproducible Financial NLP Pipeline for Post-Earnings Direction Classification: A Pre-Registered Null Result on SEC Earnings Press Releases

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Abstract

We ask whether the language of a company's earnings press release, encoded by a frozen FinBERT model, predicts the direction of its stock over the three trading sessions after the release is public. We build a leakage-safe pipeline for this question:

- an explicit information timeline in U.S. Eastern time;
- SEC EDGAR press releases (Form 8-K, Item 2.02) dated by their exact acceptance times;
- provenance-keyed caches;
- date-grouped, purged chronological splits;
- two synthetic controls that bracket the pipeline.

Selection used validation data only and was committed before the test set was evaluated, once. The positive control reaches a test AUC of 1.000 and the negative control 0.465 ± 0.075 over 5 seeds, so the pipeline transmits label signal when it exists and does not manufacture it. On 1138 real events from 30 large U.S. tickers (2016–2026), no model distinguishes the sign of the post-entry return from chance on the 229 held-out events. The FinBERT fusion head reaches a test AUC of 0.506 (95% week-block bootstrap interval 0.439–0.573), and no model beats a baseline that knows only the company's historical up-rate. We report this null result with its limits. The label measures drift after the announcement reaction, not the reaction itself.

1 Introduction

Earnings announcements are among the most information-dense events in equity markets. A large literature documents both the price response to earnings news and a slower post-announcement drift [2, 3], and another shows that the tone of financial text is related to returns [10, 13, 17, 18]. Pretrained language models specialised for finance, such as FinBERT [1], make it easy to turn text into features. This ease is also a hazard: with timestamps that are approximate, documents that appear after the fact, and news that reports the price move itself, it is easy to build a classifier that "predicts" returns by reading the future.

This report is about doing the experiment carefully rather than about a new model. FinBERT is used as published, and the classifier is deliberately small. The contributions are:

1. **An information timeline** in U.S. Eastern time. Prediction time, entry price, news cutoff and label window are defined so that no feature postdates the prediction, and the invariants are checked when the dataset is assembled (Section 4).

2. **A free, official document stream:** earnings press releases filed with the SEC as Exhibit 99.1 of Form 8-K Item 2.02 reports, dated by EDGAR acceptance times, with documented coverage gaps (Section 3).

3. **Engineering for reproducibility:**

   - explicit data modes that never fall back to synthetic data;
   - caches keyed on every input that can change their content;
   - per-run manifests;
   - frozen, fingerprinted datasets.

4. **A pre-registered evaluation** [15]:

   - synthetic positive and negative controls;
   - a ticker-identity baseline that a text model must beat before any language signal is claimed;
   - week-block bootstrap intervals, which respect the dependence between events that share market moves.

The real-data result is null. We think a clearly bounded null obtained this way is more useful than an optimistic number obtained otherwise.

2 Related work

Earnings news and prices. Ball and Brown [2] showed that accounting earnings surprises are reflected in prices around and after announcements, and Bernard and Thomas [3] documented post-earnings-announcement drift. Under market efficiency [6], public information in a large company's release should be incorporated quickly. A drift label that starts after the reaction is therefore a demanding target.

Text and returns. Tetlock [17] related media pessimism to market prices; Tetlock et al. [18] showed that negative words in firm-specific news forecast earnings and returns; Loughran and McDonald [13] showed that general-purpose sentiment dictionaries misclassify financial language and proposed finance-specific word lists; and Ke et al. [10] learned return-predictive sentiment directly from news text. Qin and Yang [16] predicted volatility from the text and audio of earnings conference calls.

Financial language models. FinBERT [1] further trains BERT [5] on financial text and fine-tunes it for three-way sentiment on the Financial PhraseBank [14]; we use the public ProsusAI checkpoint of that model. A different model with the same name, trained on corporate filings, analyst reports and call transcripts, was introduced by Huang et al. [8]. For tests we use the compact BERT of Turc et al. [19].

Methodology. Leakage, that is, information in the features that would not be available at prediction time, is a recurring cause of over-optimistic results [9]. López de Prado [12] discusses purging training observations whose label windows overlap the evaluation period, which we apply to the chronological split. For inference under dependence we use a block bootstrap [11] over calendar weeks, in the spirit of cluster bootstraps [4]. AUC is interpreted as a ranking probability [7].

3 Data

Universe and events. We requested the 40 most recent realised earnings announcements of 30 large U.S. tickers from Yahoo Finance (via `yfinance`), 1200 announcements in total, with their announcement timestamps and daily adjusted open and close prices. The universe is fixed in the code and was chosen for liquidity and coverage, not by performance.

Documents. For each announcement we take the company's earnings press release from SEC EDGAR: the Form 8-K filings whose items include 2.02 ("Results of Operations and Financial Condition"; item numbering as amended by U.S. Securities and Exchange Commission [21]), excluding amendments.

- **Exhibit choice.** The text is the exhibit typed `EX-99.1` (1131 of 1148 events). Failing that, it is `EX-99.01` or `EX-99`, then an exhibit whose index description names a press, earnings or news release, then the first `EX-99.*`.
- **Matching.** A release is matched to the announcement whose date is closest, within $[-1, +3]$ days of its ET acceptance date, and no release is used twice.
- **Access.** Requests carry the contact User-Agent required by the SEC's access policy [20].

1148 announcements yielded a usable release. Coverage gaps are structural:

- ExxonMobil filed only one Item 2.02 report since 2016, so it is effectively absent;
- Disney's filings under its current registrant begin in 2019;
- 51 announcements had no matching filing.

After purging (below), 1138 events remain, of which 684 were released after the close, 448 before the open and 6 during the session.

Document length. Releases are long: the median release has 6,862 FinBERT tokens (10th–90th percentile 2,999–12,421). 89% are longer than the 8-chunk cap used below, which encodes a median 45% of a release. Formats vary across issuers: some companies publish shareholder letters or slide-style updates as the exhibit.

Headlines. Company headlines come from Finnhub's free tier, which retains about a year of history. Only 111 events, all entering on or after 2025-11-05, have headlines. Because these all fall in the final evaluation period, headlines are excluded from the primary benchmark and analysed only in a training-free exploratory study (Section 8).

Freezing. Both collections were frozen to fingerprinted files before any model was trained on real data. Every experiment reads these files, and nothing is re-collected. The files contain provider data and are not redistributed; their SHA-256 hashes are recorded with the code.

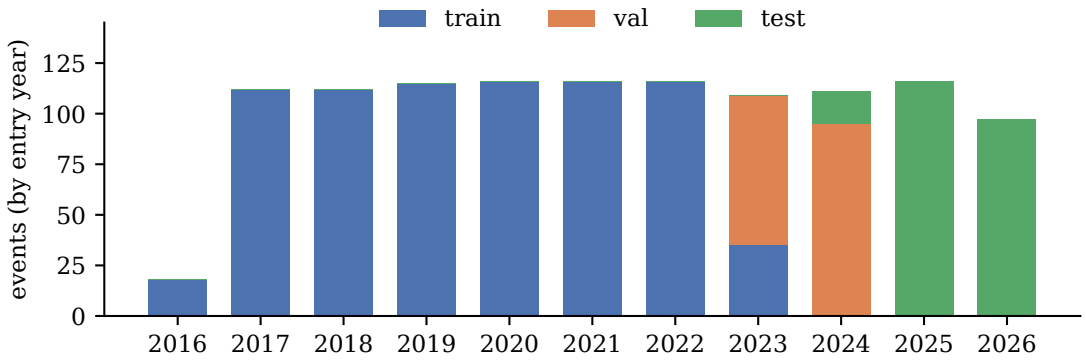


Figure 1: Events by entry year and split (document-only dataset, after purging).

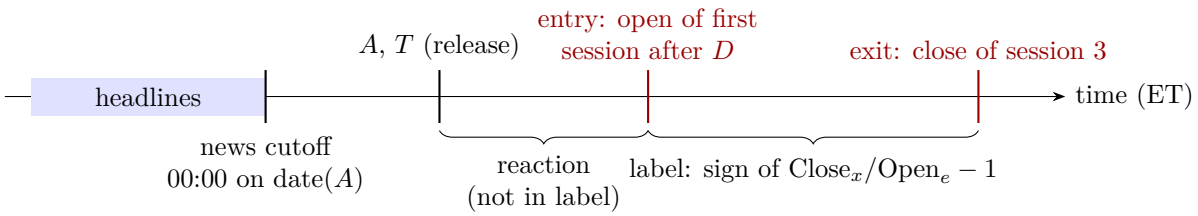


Figure 2: Information timeline for one event. Features use only text available before the entry open; the label starts after the announcement reaction.

4 Information timeline

All times are in America/New_York. Let A be the announcement time and T the document time (the EDGAR acceptance time).

- **Session.** An event is BMO if A is before 09:30, AMC if at or after 16:00, and DMH otherwise.
- **News cutoff.** Headlines count only if published before 00:00 ET on the date of A, or before T if that is earlier. Provider timestamps are approximate, so same-day news is excluded rather than trusted.
- **Information date.** $D = \max(\text{date}(A), \text{date}(T))$.
- **Prediction and entry.** The prediction is made at 09:30 ET on the first trading day strictly after D, and the position enters at that day's open. This deliberately skips a pre-open report's same-day open, because results may still be being discussed at 09:30.
- **Label.** The label is $\mathbf{1}[\text{Close}_x/\text{Open}_e > 1]$, where e is the entry day and x is the third trading session counting e as the first. Trading days come from the price series, so holidays are handled.

Each of these is re-checked on the stored data: every headline timestamp is at or before the cutoff, the entry is after the information date, and the label agrees with the stored return (Figure 2). Because entry follows the announcement, the immediate price reaction is already in the entry price. *The label measures post-announcement drift.*

5 Model and baselines

Encoder. FinBERT is used frozen, pinned to a specific checkpoint revision. A document is split into consecutive, non-overlapping 384-token chunks (≤ 8 kept); each chunk is encoded and mean-pooled over tokens, and chunk vectors are mean-pooled into one 768-dimensional document vector. The pipeline also supports a second stream for headlines. When a stream is absent, its vector is zeroed and a presence flag records the absence.

Fusion head. The head concatenates the two vectors, their absolute difference, their product and the presence flags, and applies dropout and either a single linear layer or one hidden layer. It is trained with AdamW and early stopping on validation AUC. With one stream present, the fused features reduce to the document vector and its magnitude.

Comparisons. Every model sees only release text or less:

- **Train-majority:** the training up-rate for every event.
- **Ticker prior:** each company's training up-rate, shrunk toward the global rate, $(\text{ups} + 10\,\bar{y})/(n + 10)$. Every release names its company, so the embedding encodes identity. A text model that cannot beat this prior has not shown that the language matters.
- **Logistic regression** on the same frozen document features.
- **FinBERT fusion head.**
- **Zero-shot FinBERT:** the checkpoint's own sentiment classifier, scoring each of the first 8 chunks as $P(\text{positive}) - P(\text{negative})$ and averaging. Nothing is trained.

6 Protocol

The protocol was written after the data were collected and before any model was trained on real data, and was committed to version control. Its decisions are listed below.

Pre-registration here means the protocol was committed to version control before any real-data training, and the selection file before the test stage. The public repository starts from a single commit, so this order is documented in the protocol's change log and in the development-history commit identifiers stored in the result files rather than in the public history; it was not lodged with an external registry.

- **Split.** Events are ordered by entry date and split chronologically at date boundaries (Table 1). Train events whose label windows reach into validation, and validation events reaching into test, are purged (6 and 4 events).
- **Selection.** On validation only:
  - logistic regression over 1 or 8 chunks $\times$ $C \in \{0.01, 0.1, 1\}$;
  - the fusion head over 1 or 8 chunks $\times$ 0 or 1 hidden layers, each with 5 seeds, choosing the best mean validation AUC.

  The selected configurations were committed before the test stage, which refuses to run unless that file matches the committed version.

Table 1: Chronological split of the document-only dataset (after purging).

Split	First entry	Last entry	n	Up-rate
Train	2016-10-19	2023-04-24	740	0.470
Val	2023-04-27	2024-10-24	169	0.527
Test	2024-10-31	2026-10-23	229	0.454

Table 2: Synthetic pipeline controls (FinBERT, seeds 0–4). Not market evidence.

Control	Runs	Head test AUC	Range	LR test AUC	Head val AUC
Negative	5	0.465 ± 0.075	0.392–0.578	0.469	0.548
Positive	5	1.000 ± 0.000	1.000–1.000	1.000	1.000

- **Test.** The test set was evaluated once.
- **Inference.**
  - The primary metric is ROC AUC; AP, Brier score, log loss, accuracy and F1 are reported alongside it.
  - Intervals come from 2,000 bootstrap resamples of whole entry weeks: events in the same week share market-wide moves.
  - Model comparisons use paired differences on the same resamples.
- **Claim rule, fixed in advance.** Release-language signal may be claimed only if a FinBERT model beats the ticker prior with a paired 95% interval excluding zero.

7 Synthetic controls

Two synthetic datasets with the real schema and timeline bracket the pipeline. The "documents" are built from stock sentences.

- **Negative control.** Labels are drawn independently of the text, so any AUC far from 0.5 would indicate leakage.
- **Positive control.** The text is bullish or bearish according to the label, and long enough to span several chunks, so a working pipeline must separate the classes.

With FinBERT and 5 seeds each (about 79 test events per seed), the negative control scores a test AUC of 0.465 ± 0.075 (range 0.392–0.578), consistent with chance at this sample size. The positive control scores 1.000 on every seed (Table 2, Figure 3). These are pipeline checks, not market evidence.

8 Real-data results

Selection (validation). Table 3 shows the validation grid. Logistic regression was selected with 1 chunk and $C = 0.01$; the fusion head with 1 chunk and 0 hidden layers (mean validation AUC 0.602). Configurations using 8 chunks did not do better on validation.

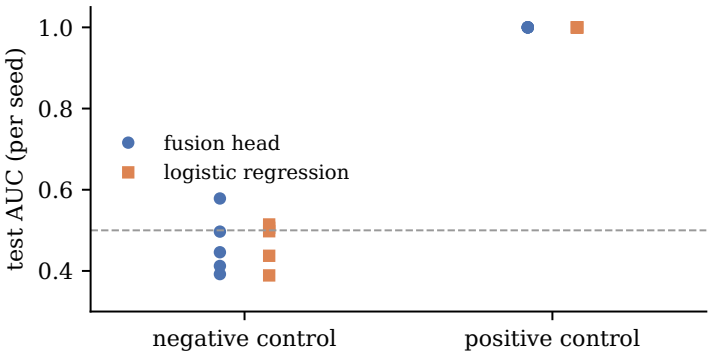


Figure 3: Per-seed test AUC of the synthetic controls.

Table 3: Validation grid used for selection (validation data only). Fusion-head values are mean ± s.d. over 5 seeds; each seed's value is at its early-stopping epoch.

Model	Setting	Chunks	Val AUC
Logistic regression	$C = 0.01$	1	0.539
Logistic regression	$C = 0.1$	1	0.523
Logistic regression	$C = 1$	1	0.524
Logistic regression	$C = 0.01$	8	0.498
Logistic regression	$C = 0.1$	8	0.453
Logistic regression	$C = 1$	8	0.452
Fusion head	0 hidden layers	1	0.602 ± 0.008
Fusion head	1 hidden layer	1	0.594 ± 0.005
Fusion head	0 hidden layers	8	0.568 ± 0.010
Fusion head	1 hidden layer	8	0.579 ± 0.007

Held-out test. On 229 test events in 79 entry weeks (2024-10-31 to 2026-10-23), no model is distinguishable from chance (Table 4, Figure 4).

- The fusion head reaches 0.506 (0.439–0.573); its seeds range 0.497–0.516.
- Logistic regression reaches 0.461, zero-shot FinBERT 0.469 and the ticker prior 0.487.
- The paired difference between the fusion head and the ticker prior is +0.019 (−0.052 to +0.089; Table 5). By the pre-registered rule, no release-language signal is claimed.
- On probability quality, the fusion head's Brier score (0.259) and log loss (0.711) are no better than the train-majority's (0.248, 0.689).

Validation optimism. Validation AUCs of 0.532–0.601 did not carry over to test, for two identifiable reasons:

- The fusion head's validation number is optimistic by construction, because each seed is stopped at its best validation epoch.

Table 4: Held-out test results (evaluated once). Intervals: 95% week-block bootstrap. Zero-shot FinBERT produces rank-only scores, so probability metrics are not reported for it.

Model	Val AUC	Test AUC	95% CI	AP	Brier	Log loss	Acc.	F1
Train-majority	0.500	0.500	[0.500, 0.500]	0.454	0.248	0.689	0.546	0.000
Ticker prior	0.566	0.487	[0.423, 0.556]	0.448	0.258	0.710	0.528	0.407
Logistic regression	0.539	0.461	[0.383, 0.538]	0.456	0.293	0.786	0.450	0.515
FinBERT fusion head	0.601	0.506	[0.439, 0.573]	0.455	0.259	0.711	0.463	0.580
Zero-shot FinBERT	0.532	0.469	[0.388, 0.552]	0.434	–	–	–	–

Table 5: Paired test-AUC differences on the same week-block resamples, with the share of resamples in which the difference is ≤ 0.

Test AUC difference	Estimate	95% CI	Share ≤ 0
FinBERT fusion head − Ticker prior	+0.019	[−0.052, +0.089]	0.31
Logistic regression − Ticker prior	−0.026	[−0.111, +0.052]	0.73
Zero-shot FinBERT − Ticker prior	−0.018	[−0.119, +0.081]	0.66
FinBERT fusion head − Logistic regression	+0.045	[−0.042, +0.129]	0.16
Ticker prior − Train-majority	−0.013	[−0.077, +0.056]	0.64

- The ticker prior also fell from 0.566 on validation to 0.487 on test: the companies that had tended to rise after earnings changed between periods.

Selecting on the validation set and evaluating once on a later test set is designed to expose exactly this kind of gap.

Exploratory: headlines. Labelled exploratory; nothing was trained or selected. On the 111 events that have pre-cutoff headlines (39 entry weeks), zero-shot FinBERT sentiment of the headlines has an AUC of 0.429 (0.323–0.556). On the 109 of those events that also have a release, release sentiment scores 0.413, and the difference is +0.022 (−0.127 to +0.182). Neither score distinguishes direction from chance in this small sample.

9 Limitations and threats to validity

- **Target.** The label is three-session drift after the entry open, not the announcement reaction, which is already priced in for these companies. The result says nothing about whether release text explains the reaction.
- **Precision.** The test interval half-width is about 0.067 AUC. Effects smaller than that cannot be detected here.
- **Text coverage.** Only the first ∼380 tokens (selected) or ∼3k tokens of releases with a median of 6,862 tokens are encoded. Call transcripts, analyst Q&A and expectations (the surprise) are not used, and the encoder is not fine-tuned.
- **Universe.** The universe is today's large capitalisation names projected back to 2016, which introduces survivorship bias. ExxonMobil is effectively absent and Disney is missing before 2019. Results may differ for smaller, less-covered firms.

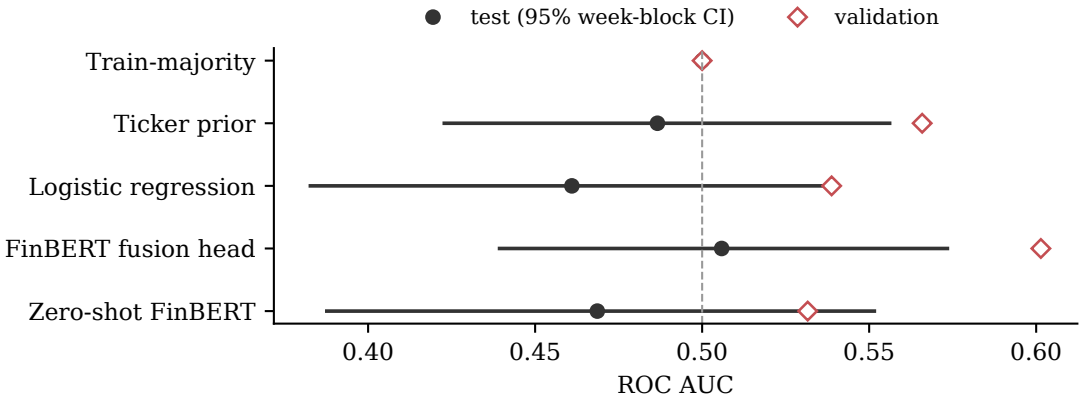


Figure 4: Validation AUC (open diamonds) and held-out test AUC with 95% week-block intervals.

- **Timing data.** Yahoo announcement times are approximate. The conservative cutoff and the entry rule make the label insensitive to the stamped minute, at the cost of excluding same-day news.
- **Identity.** Releases name their company, so embeddings encode identity. The ticker prior controls for this, but only for stable company effects.
- **Reproduction.** The frozen datasets contain provider data and are not redistributed. A fresh collection with the same code yields a similar but not identical dataset, because provider data change. The code, fingerprints, manifests and all result files are public.
- **Not trading performance.** Classification metrics are not trading returns, and nothing here supports a trading claim.

10 Reproducibility

The repository contains the code, tests, dependency lock, pre-registered protocol, run manifests and the result files from which every number, table and figure in this paper is generated by a script (`experiments/paper_assets.py`). The real-data steps are:

```
python main.py --data-mode real --collect-only   # collection + coverage
python experiments/phase4.py select              # validation only
python experiments/phase4.py test                # needs committed selection
python experiments/phase4.py headlines           # exploratory
python experiments/paper_assets.py               # numbers, tables, figures
python main.py --data-mode synthetic-positive-control   # no credentials
```

The synthetic controls need no credentials. Real collection needs a contact User-Agent for SEC EDGAR and, for headlines, a Finnhub key, both supplied through environment variables.

References

[1] Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models, 2019. URL https://arxiv.org/abs/1908.10063.

[2] Ray Ball and Philip Brown. An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2):159, 1968. doi: 10.2307/2490232.

[3] Victor L. Bernard and Jacob K. Thomas. Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27:1, 1989. doi: 10.2307/2491062.

[4] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

[5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, 2019. doi: 10.18653/v1/N19-1423.

[6] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383, 1970. doi: 10.2307/2325486.

[7] James A. Hanley and Barbara J. McNeil. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1):29–36, 1982. doi: 10.1148/radiology.143.1.7063747.

[8] Allen H. Huang, Hui Wang, and Yi Yang. FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2):806–841, 2023. doi: 10.1111/1911-3846.12832.

[9] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012. doi: 10.1145/2382577.2382579.

[10] Zheng Tracy Ke, Bryan T. Kelly, and Dacheng Xiu. Predicting returns with text data. NBER Working Paper 26186, National Bureau of Economic Research, 2019. URL https://www.nber.org/papers/w26186.

[11] Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989. doi: 10.1214/aos/1176347265.

[12] Marcos López de Prado. *Advances in Financial Machine Learning.* Wiley, 2018. ISBN 9781119482086.

[13] Tim Loughran and Bill McDonald. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1):35–65, 2011. doi: 10.1111/j.1540-6261.2010.01625.x.

[14] Pekka Malo, Ankur Sinha, Pekka Korhonen, Jyrki Wallenius, and Pyry Takala. Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*, 65(4):782–796, 2014. doi: 10.1002/asi.23062.

[15] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

[16] Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 390–401, 2019. doi: 10.18653/v1/P19-1038.

[17] Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168, 2007. doi: 10.1111/j.1540-6261.2007.01232.x.

[18] Paul C. Tetlock, Maytal Saar-Tsechansky, and Sofus Macskassy. More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance*, 63(3):1437–1467, 2008. doi: 10.1111/j.1540-6261.2008.01362.x.

[19] Iulia Turc, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Well-read students learn better: On the importance of pre-training compact models, 2019. URL https://arxiv.org/abs/1908.08962.

[20] U.S. Securities and Exchange Commission. Accessing EDGAR data. https://www.sec.gov/os/accessing-edgar-data. Accessed 2026-10-27.

[21] U.S. Securities and Exchange Commission. Additional form 8-k disclosure requirements and acceleration of filing date. Final rule, 69 FR 15593; effective August 23, 2004, 2004. URL https://www.sec.gov/rules/final/33-8400.htm.